Fiscal Year:	FY 2011	**Task Last Updated:**	FY 04/24/2011
PI Name:	Narayanan, Vinod Ph.D.		
Project Title:	Boiling Heat Transfer Mechanism During Thermally Actuated Pumping by Asymmetric Surface Structures		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	vinod.narayanan@oregonstate.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	541-737-7012
Organization Name:	Oregon State University		
PI Address 1:	Mechanical Industrial and Manufacturing Engineering		
PI Address 2:	204 Rogers Hall		
PI Web Page:			
City:	Corvallis	**State:**	OR
Zip Code:	97331	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2009	**End Date:**	09/30/2012
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
Contact Email:	david.f.chao@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date is 9/30/2012 per NSSC information (Ed., 2/28/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bhavnani, Sushil (Co-PI: Auburn University)		
Grant/Contract No.:	NNX09AJ98G (OSU) ; NNX09AL63G (Auburn)		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Proposed is a collaborative research study between investigators at Oregon State University (OSU) and Auburn University (AU). A novel thermally actuated pumping mechanism during boiling heat transfer is identified in the context of passive means of enhancing heat transfer rate. The effect of an asymmetric force on the boiling process is the focus of the study. Such asymmetry is created by the use of a surface with repeated ratchet structures and pyramidal reentrant cavities located on one of its sides. The premise of the proposal is that surface asymmetry results in a force asymmetry during the growth and departure of the vapor bubbles causing local fluid motion in the streamwise direction (parallel to the surface). Forces that cause this fluid motion will be quantified at the microscale using minimally intrusive imaging methods. The hypothesis is that with several ratchets, this local motion can be translated to a net fluid pumping. The project has the following major tasks: 1) Fabricate ratchet –cavity structures surfaces using silicon microfabrication methods, 2) Perform proof-of-principle experiment, using high-speed flow visualization, confirming the bulk movement of the fluid by the proposed mechanism, 3) Measure, by analysis of High-Speed (HS) flow visualization movies, the time-varying bubble diameters and frequencies, 4) Determine, by temporally- and spatially- resolved temperature and heat flux measurements, heat transfer rate from the surface to the fluid in the small array ratchet experiments, and 5) Perform global measurements in the large array experiments. Program Significance: Passive methods of heat transfer enhancement are attractive for space and terrestrial applications since they use no external power and need little maintenance. The proposed phase-change enhancement technique can be applied for space thermal management systems and for enhancing heat transfer rate from adversely oriented surfaces. In microgravity pool boiling, it has been shown that the presence of a large primary vapor bubble near the heater surface causes premature dryout. Axial motion of this bubble by force asymmetry of the proposed concept could not only prevent local dryout but also enhance heat transfer. The proposed imaging methods present a detailed estimation of flow and temperature field in boiling flows that are of use to modellers. The proposed experiments and analyses will highlight forces of importance by varying orientations, subcooling and gravity levels, and use of fluids with contrasting properties. Broader Significance: Two graduate students will gain experience in the field of importance to NASA. Results from the project will be integrated in the form of a module into an existing Electronics Thermal Management live-internet multi-campus course at AU and will incorporate OSU as a new partner. Undergraduate students from both AU and OSU will be involved in the DC-9 flight experiments along with graduate students. The Oregon NASA Space Grant Consortium will help provide travel support for the DC-9 experiments as well as provide an avenue for dissemination of results to the community.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is the second year of the grant. Proof of concept high-speed visualization results have confirmed that bubbles emanating from cavities located on the face of an inclined surface have an axial component of velocity to them. Furthermore, under sub-cooled condition of 20 degrees C for water, bubble collapse due to condensation significant increases the bubble velocity as well as the axial component of this velocity. The demonstration of axial motion of bubbles leads to the possibility of thermally actuated pumping of fluid. Since the mechanism of pumping occurs in the nucleate boiling regime, this discovery opens up the possibility of creating thermal loops capable of dissipating high heat fluxes passively.
Task Progress:	The following research activities have been performed over the past year of the project period: 1) Updates of the imaging test facility (OSU), 2) Completion of the large-array test facility (Auburn), 3) Fabrication of imaging and large array silicon test sections (Auburn), 4) Qualitative proof-of-concept experiments on the imaging test section (OSU), 5) Quantification of bubble velocities and accelerations in the imaging test section (OSU). Several additions were made to the imaging test facility to target future parabolic flight experiments. The multi-step silicon fabrication of the test surface was completed successfully. Proof of principle experiments have revealed that bubbles emanating from cavities of a particular shape that are located on the inclined faces of ratchets have an axial component of velocity to them. A bubble tracking algorithm was written to accurately determine the velocities and accelerations of bubbles from high-speed images. Closed loop measurements using a large array of ratchets are to be performed in the near future to determine whether the bubble axial motion translates to significant bulk motion of the fluid. Two graduate students, one at Oregon State University (OSU) and another at Auburn University (AU) are involved in the project. As a part of integrating research with education, an experiment on thermally actuated pumping in the Leidenfrost regime was introduced in ME552 Measurements in Fluid Mechanics and Heat Transfer class at Oregon State University. A new course, ME549 Special Topics in Heat Transfer- Thermophysics of Phase Change Heat Transfer, is currently being taught by the OSU PI. The Auburn PI will be presenting a lecture via videoconference as a part of this course. One conference publication has been accepted in its final form for presentation at the InterPACK conference in July 2011.
Bibliography Type:	Description: (Last Updated: 02/03/2014)
Papers from Meeting Proceedings	Thiagarajan N, Kapsenberg F, Narayanan V, Bhavnani S. "On The Lateral Motion of Bubbles Generated from Re-Entrant Cavities Located on Asymmetrically Structures Surfaces." ASME 2011 Pacific Rim Technical Conference and Exposition on Packaging and Integration of Electronic and Photonic Systems, InterPACK, Portland, OR, July 6-8 2011. Proceedings of the ASME 2011 Pacific Rim Technical Conference and Exposition on Packaging and Integration of Electronic and Photonic Systems, InterPACK, Portland, OR, July 6-8 2011. InterPACK2011-52056. In press, April 2011. http://www.asmeconferences.org/InterPACK2011/index.cfm , Apr-2011